Customer Support System - Key Features & Technical Capabilities

The Customer Support System is a professional-grade, multi-application infrastructure designed to provide high-performance real-time support. Below are the core technical features that differentiate this platform.

1. Intelligent Agent Assignment

The system features an automated ticket routing engine that ensures optimal workload distribution. - **Least Busy Algorithm**: Automatically assigns incoming chats to agents with the lowest current workload. - **Dynamic Reassignment**: If an agent changes status to "Away" or disconnects, active chats are automatically returned to a "Pending" state and queued for the next available agent. - **Max Load Enforcement**: Prevents agent burnout by limiting the number of simultaneous active chats per agent.

2. Real-Time Communication Ecosystem

Built on top of Socket.IO, the system provides a seamless, low-latency communication experience. - **Sub-50ms Latency**: Real-time message delivery across all connected clients. - **Typing Indicators**: Live visual feedback for both customers and agents. - **Presence Tracking**: Real-time monitoring of agent availability (Online, Busy, Away).

3. Comprehensive Analytics and Logging

Every interaction within the system is logged and indexed for auditing and performance analysis. - **Activity Streams**: Full history of agent status changes and ticket transitions. - **Message Histories**: Persistent storage for all customer-agent conversations. - **Dashboard Metrics**: Real-time stats for administrators to monitor system health and response times.

4. Scalable Micro-App Architecture

The platform is composed of 7 distinct applications, allowing for granular scaling and maintenance. - **Decoupled APIs**: Independent Node.js backends for e-commerce and support functions. - **Fragmented Frontends**: Specialized dashboards for agents and admins using React and Angular, respectively. - **Seamless Integration**: A plug-and-play chat widget that can be embedded into any external web property.

5. Security and Access Control

Robust security measures protect customer data and system integrity. - **JWT Authentication**: Token-based security for all API and WebSocket transactions.

- **Role-Based Access Control (RBAC)**: Detailed permission sets for Admins, Agents, and Customers. - **Email Verification**: Multi-step registration process to ensure valid user identities.